

When a Number Rule Stops Working

Directions

- Each problem shows someone who looked at a few numbers and decided on a rule.
- A rule that says *every* has to survive *every* case. One number that breaks it is enough.
- Write the number the question asks for, then answer the explanation part in full sentences.

1. Ravi is writing a pattern that starts at 4 and adds 6 each time. He wrote:

4, 10, 16, 24, 30

- (a) Write the correct 4th term.

Answer: _____

- (b) Write the correct 5th term.

Answer: _____

- (c) Explain what Ravi did wrong.
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2. Mia says: “Every whole number that ends in the digit 7 is prime.” Find the smallest number greater than 7 that ends in 7 and is *not* prime, and write it as a product of two whole numbers, neither of them 1.

Answer: _____

3. Ben writes a pattern that starts at 3 and adds 4 each time. He checks the first three terms, sees that each one is prime, and says: “Every number in my pattern is prime.”

- (a) Write the first term of Ben’s pattern that is *not* prime.

Answer: _____

- (b) Explain why checking three terms was not enough.

4. Omar says: “Every multiple of 3 is odd. Look at 3, 9, 15 and 21.”

- (a) Write the smallest multiple of 3 that is greater than 3 and is even.

Answer: _____

- (b) Explain why Omar’s examples led him to a rule that is false.

5. A pattern begins 1, 4, 9, 16, ... Toby sees that the first step is +3 and decides the rule is “add 3 every time”. He writes 13 as the 5th term.

(a) Write the correct 5th term.

Answer: _____

(b) Explain why Toby’s rule does not fit this pattern.

6. Priya says: “A pattern that starts with a bigger number always has bigger terms later on.” Pattern A starts at 5 and adds 3 each time. Pattern B starts at 2 and adds 6 each time.

(a) Write the 10th term of pattern A.

Answer: _____

(b) Write the 10th term of pattern B.

Answer: _____

(c) Write $<$ or $>$ between your two answers.

Answer: _____

(d) Explain why Priya’s rule is wrong.

7. A pattern starts at 1 and each term is 3 times the term before it. Dana says: “Every term is odd, so the rule must be add an even number each time.”

(a) Write the 6th term of the pattern.

Answer: _____

(b) Explain why Dana’s rule cannot be the rule for this pattern.

8. Ines says: “A number divisible by 2 and by 3 is always divisible by 6. So a number divisible by 2 and by 4 must always be divisible by 8.”

(a) Write the smallest whole number greater than 8 that is divisible by both 2 and 4 but is *not* divisible by 8.

Answer: _____

(b) Explain why the first rule works but the second one does not.
